

S-Plane detection — audit, corrections, and final validated results

Date: 2026-09-20 (overnight) · **Status:** corrected and re-validated · **Supersedes** all earlier 2026-09-20 result sheets

Why this document exists

A cross-check audit of the same day's work found **four defects, two of them fabricating data**. Everything below is the corrected position. Earlier numbers issued on 2026-09-20 (v2 sensitivity 1.000 [0.962,1.000], specificity 0.800, C1/C2/C3 12/12) are **withdrawn**. Nothing is hidden; each defect is stated with the evidence that proved it.

The four defects

D1 — 36 "replicates" were one capture copied 12 times (*fabricated data*)

`run_gap.sh` was created mode **644** but invoked as `./run/run_gap.sh`, returning `**exit 126`

(Permission denied), **silenced by** `>/dev/null 2>&1`. The packet-count guard then read a stale** capture left in the working directory, passed it, and copied it into all 12 replicate folders.

Proof: all 12 captures per C-scenario had **one distinct SHA-256**; the three scenarios completed in **≤1 second** per rep in the campaign log, where a genuine run takes ~48 s.

Fix: permissions corrected; runner now invoked via `bash`, **exit code checked**, and the working directory **purged before every run** so a stale capture can never satisfy the guard. All 36 runs re-executed.

D2 — B3_pdv_congestion never injected anything (*fabricated scenario, also affects the earlier campaign*)

This kernel has no `sch_netem` (`tc ... netem → "Specified qdisc kind is unknown"`), and `scenarios.sh` swallowed the failure with `2>/dev/null||true`. B3 was a **baseline run wearing a congestion label**.

Proof: B3 PDV was statistically identical to baseline (ru1 median 62.4 ms / sd 36.0 ms vs 61.7 / 35.5), and identical to the untouched ru3 control.

Fix: replaced with a **physical congestion model** — a `tbfb` bottleneck on the BC → RU timing path plus competing **non-PTP** background traffic through it, so PTP queues behind data-plane traffic. Deep queue ⇒ packets are *delayed, not dropped*. Sync PDV now rises from **0.07 ms to 5.6–857 ms**. All 12 reps re-run. **This defect also invalidates B3 in the earlier 132-run campaign cited in the submitted documents.**

D3 — the v2 detector produced false positives and destroyed an honest abstention

v2's rate-starvation detector (D1) had **no minimum-sample and no transient guard**. A boundary clock that

merely appears briefly during BMCA start-up (3–5 frames) was read as "rate starved."

Proof (upstream captures): B2_gm_failover **BENIGN** → **ATTACK**; B_unplanned_failover **UNKNOWN** → **ATTACK**,

i.e. it overwrote the deliberate abstention. Trigger: 3 frames → 0.06 Hz vs 16 Hz declared.

This is the exact failure the frozen base rule had already solved with TRANSIENT_FRACTION; v2 reintroduced it.

Fix: decision_rule_v3.py (frozen aa9417b7) adds a minimum-frame floor and a persistence requirement (a source must be active across ≥50 % of the window before its rate is judged), includes the provisioned replacement BC identity, and requires a UTC-offset change to be sustained. v3 fixes both false positives and is **verdict-identical to v2 on every downstream capture** — it removes false positives without altering a single true positive.

D4 — the evaluator overstated and under-checked

evaluate_v3.py pooled metrics in a way that moved with run count, "proved" additive-only by comparing two scalars, credited abstention even on error paths, had no replicate whitelist, and silently dropped attribution scoring. Replaced by evaluate_v4.py: per-scenario primary metrics, macro-averages, per-run additive-only verification, substantive-path-only abstention, explicit rep whitelist, restored attribution, and a fully enumerated run list written into the artifact.

Disclosure — frozen-manifest changes

run/randparams.py and run/scenarios.sh no longer match the 2026-09-17 manifest (verify_frozen.sh will flag them). Reasons are recorded in FROZEN_HARNESS_EPOCH2.json. **The decision rules themselves are unchanged and still match their original hashes** — only the experiment apparatus changed.

Corrected results — 168 live runs (12 reps × 14 scenarios), all three rules on the same runs

Per-scenario is the **primary** metric. Pooled figures are descriptive only: because both rules are bimodal, a pooled number moves with how many runs of each scenario exist, not with rule quality.

Scenario	Expect	Frozen rule	v3 (primary)	95% CI (v3)
A1 rogue master	ATT ACK	12/12	12/12	[0.757, 1.000]
A2 sync spoof	ATT ACK	12/12	12/12	[0.757, 1.000]
A3 replay	ATT	12/12	12/12	[0.757,

	ACK			1.000]
A5 DoS flood	ATT ACK	12/12	12/12	[0.757, 1.000]
A8 rogue BC	ATT ACK	12/12	12/12	[0.757, 1.000]
C1 packet removal	ATT ACK	0/12	11/12	[0.646, 0.985]
C2 malformed	ATT ACK	0/12	12/12	[0.757, 1.000]
C3 whole-second	ATT ACK	0/12	12/12	[0.757, 1.000]
baseline	BENI GN	12/12	12/12	[0.757, 1.000]
B2 GM failover	BENI GN	12/12	12/12	[0.757, 1.000]
B3 congestion <i>(now real)</i>	BENI GN	11/12	11/12	[0.646, 0.985]
B7 topology change	BENI GN	12/12	12/12	[0.757, 1.000]
B_bc_replaceme nt	BENI GN	0/12	0/12	[0.000, 0.242]
B_unplanned_fail over	UNK NOW N	12/12	12/12	[0.757, 1.000]

Summary	Frozen rule	v3 (primary)
Macro sensitivity (8 attack scenarios)	0.625	0.990
Macro specificity (5 benign scenarios)	0.783	0.783 (identical)

Abstention correctness (substantive path)	1.000	1.000
Attribution accuracy	0.552	0.875

Additive-only, verified per run (not by scalar): 0 runs where the frozen rule said ATTACK and v3 did not.
35 escalations, all on C1/C2/C3. v2 and v3 agree on every downstream run.

Measured sensitivity limit (this is a real result, not a caveat)

C1 is **11/12**, and the miss is precisely located. D1 fires when delivered rate drops below 0.5× declared:

blackhole	observed/ declared	verdict
75 %	0.366	caught
70 %	0.414	caught
65 %	0.458–0.462	caught
60 %	0.505	missed (threshold 0.500)

D1 detects sustained interception that blackholes ≥ 65 % of the observation window and evades below ~62 %.

Short or bursty interception averages out and is not detected.

Known limitations, not fixed

1. **B_bc_replacement 0/12 for both rules** — a legitimate, planned boundary-clock swap is called an attack.

Context carries a single `expected_bc_identity`. It is a deterministic known gap, not a sampling rate;

it is the entire reason specificity is 0.783 rather than 1.000. Fixing it needs a BC allow-list + re-freeze.

1. **D3 clause (c) will false-positive on real PRTC hardware** — it flags `clockClass ≤ 6` with traceability

deasserted, which is normal for many production grandmasters. Not exercised here (testbed GMs advertise 248).

1. **TRUE_TAI_UTC_OFFSET = 37** is correct only until the next leap second; override via `ctx['true_utc_offset']`.
2. **C-scenario design space is small** — randomisation draws from only 4–5 distinct values, so

12 replicates

are 12 genuine captures but not 12 independent attack *designs*.

1. **Hardware-only, out of software scope:** GNSS spoof (A6), GNSS jam (A7), delay-attack measurement (A4),

holdover (B1), SyncE/EEC (B4), oscillator drift (B6).

Evidence

splane_campaign_CORRECTED_2026-09-20.tgz (1096 files: all 168 runs' deep CSVs, three rules' verdicts per run, injection/impairment logs, every script, all freeze records) · EVALUATION_V4.json (fully enumerated run list)

· FROZEN_HARNESS_EPOCH2.json (disclosed manifest changes) · FROZEN_V3.json (why v3 exists, stated openly).

Reproduce: `python3 run/evaluate_v4.py`.